Final Project Report – Capstone 2

Traffic Accident Predicting

U.S. Accidents Dataset (2016–2023)

Student Name: Firaol Kumsa

Date: 2026

Abstract

Traffic accidents remain a significant public safety concern across the United States. This study applies end-to-end data science methodology to predict accident severity using the U.S. Accidents dataset (2016–2023). The project includes comprehensive data wrangling, feature engineering, exploratory data analysis, and the development of multiple machine learning models including Logistic Regression, K-Nearest Neighbors, Random Forest, and Gradient Boosting. Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics. Gradient Boosting achieved the highest predictive performance with 0.82 accuracy and 0.90 ROC-AUC. The results demonstrate the effectiveness of ensemble tree-based models for handling nonlinear relationships and class imbalance in real-world traffic accident data.

1. Problem Identification

Traffic accidents are a major public safety issue, causing thousands of injuries and fatalities each year. Understanding the factors that contribute to accident severity can help inform policy, improve road safety, and assist emergency response planning. This project aims to predict the severity of traffic accidents based on environmental and situational factors such as weather conditions, time of day, and location.

Project objective:

The goal of this project is to predict accident severity using environmental, temporal, and roadway-related features from the U.S. Accidents dataset (2016–2023). By building predictive models, the project aims to identify patterns and key contributing factors associated with severe crashes.

2. Data Description

The U.S. Department of Transportation and related open data portals (such as Kaggle' s U.S. Accidents dataset US Accidents (2016 - 2023)) provide detailed traffic crash records from across the country. The dataset typically includes:

- Date and time of accident
- Location features (city, state, coordinates)
- Weather conditions
- Road and traffic indicators
- Accident severity levels

The target variable is **Severity**, a categorical measure representing the seriousness of the accident.

3. Methods

3.1 Data Wrangling

Data wrangling was a critical phase of this project because the U.S. Accidents dataset is large, complex, and contains missing, inconsistent, and high-dimensional data. Proper cleaning and transformation ensured the dataset was reliable and suitable for machine learning modeling.

3.1.1 Data Inspection and Initial Cleaning

The first step involved exploring the dataset structure and identifying potential quality issues.

Actions performed:

- Reviewed dataset shape (rows and columns).
- Checked data types for each feature.
- Identified duplicate records.
- Examined missing value percentages per column.

Duplicate entries were removed to prevent biased model training. Columns with extremely high missing rates were evaluated and dropped when they did not contribute meaningful information.

3.1.2 Handling Missing Values

Missing data was present in several environmental and road-condition variables. A structured approach was used:

Numerical features

- Missing values were filled using:
    - Median values for skewed distributions
    - Mean values for normally distributed features

Categorical features

- Missing entries were replaced with:
    - "Unknown" category where appropriate
    - Mode (most frequent value) when meaningful

Feature removal

- Columns with excessive missing values and low relevance were removed to reduce noise and computational load.

This approach preserved useful information while minimizing bias introduced by missing data.

3.1.3 Data Type Conversion

Several features required transformation into appropriate formats:

- Converted date/time column to datetime format.
- Converted Boolean indicators into binary numeric format (0/1).
- Ensured severity variable was treated as a categorical/target label.

This allowed accurate time-based feature extraction and proper model compatibility.

3.1.4 Feature Engineering

To improve predictive power, new variables were derived from existing columns.

Time-based features extracted:

- Hour of day
- Day of week
- Month
- Year
- Weekend indicator
- Night vs Day indicator

These variables capture traffic behavior patterns such as rush hours, weekend travel, and nighttime driving risk.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to better understand the structure of the U.S. Accidents dataset, identify patterns, detect anomalies, and explore relationships between features and accident severity. This step helped guide feature selection, engineering, and model development.

4.1 Objective of EDA

The main goals of the exploratory analysis were to:

- Understand the distribution of accident severity levels.
- Identify important predictors influencing accident severity.
- Detect trends based on time, weather, and road conditions.
- Examine relationships among numerical variables.
- Discover potential class imbalance issues.
- Generate insights that support feature engineering and model building.

4.2 Dataset Overview

Initial exploration included:

- Examining the number of records and features.
- Reviewing summary statistics of numerical variables.
- Inspecting categorical value distributions.

- Checking class balance for the target variable (Severity).

4.3 Target Variable Analysis – Accident Severity Distribution

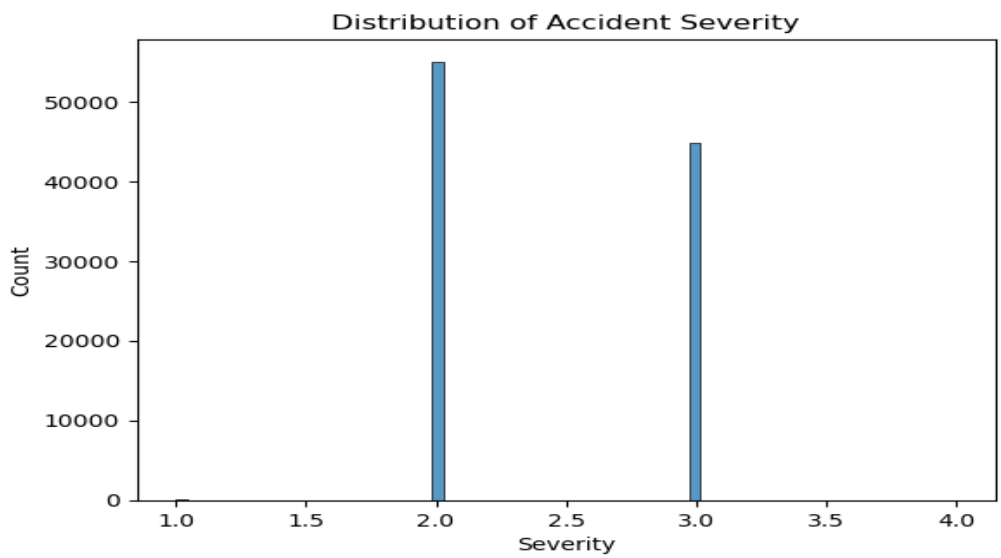
A bar chart was created to visualize the distribution of accident severity levels.

Insights:

Most accidents fall into lower severity categories.

Severe accidents occur less frequently but are critically important to predict.

The imbalance suggests the need for careful model evaluation using precision, recall, and F1-score rather than accuracy alone.



This analysis helped in understanding model expectations and evaluation strategy.

Time-related features were explored to identify patterns in accident occurrence.

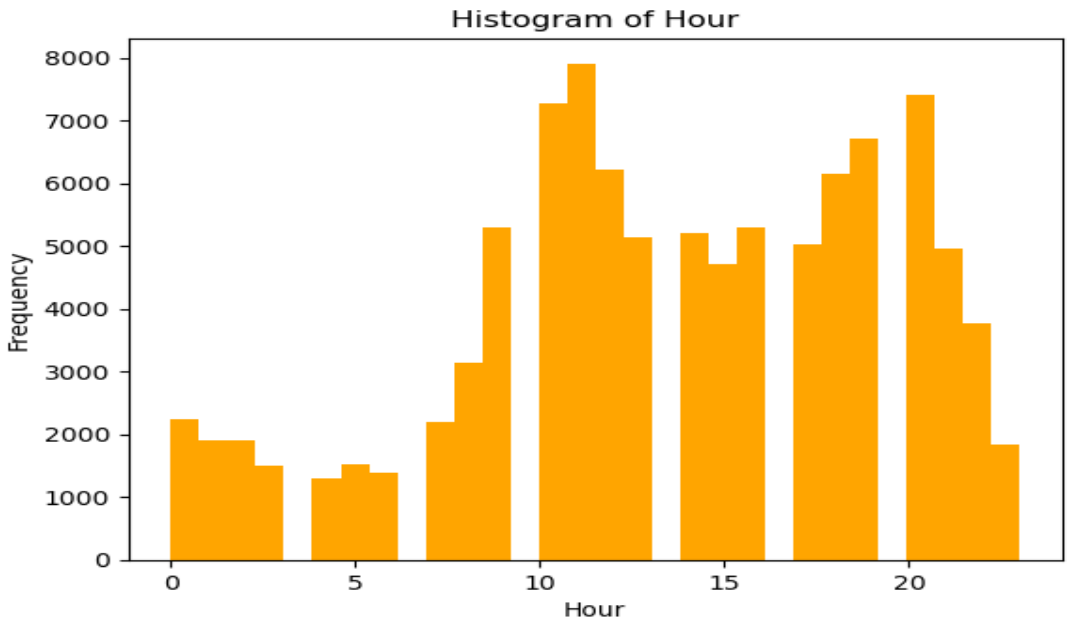
4.3.1 Accidents by Hour of Day

A histogram chart showed the frequency of accidents across hours.

Key findings:

- Higher accident frequency during rush hours (morning and evening).
- Lower frequency late at night.

Nighttime accidents tend to be more severe due to reduced visibility and fatigue

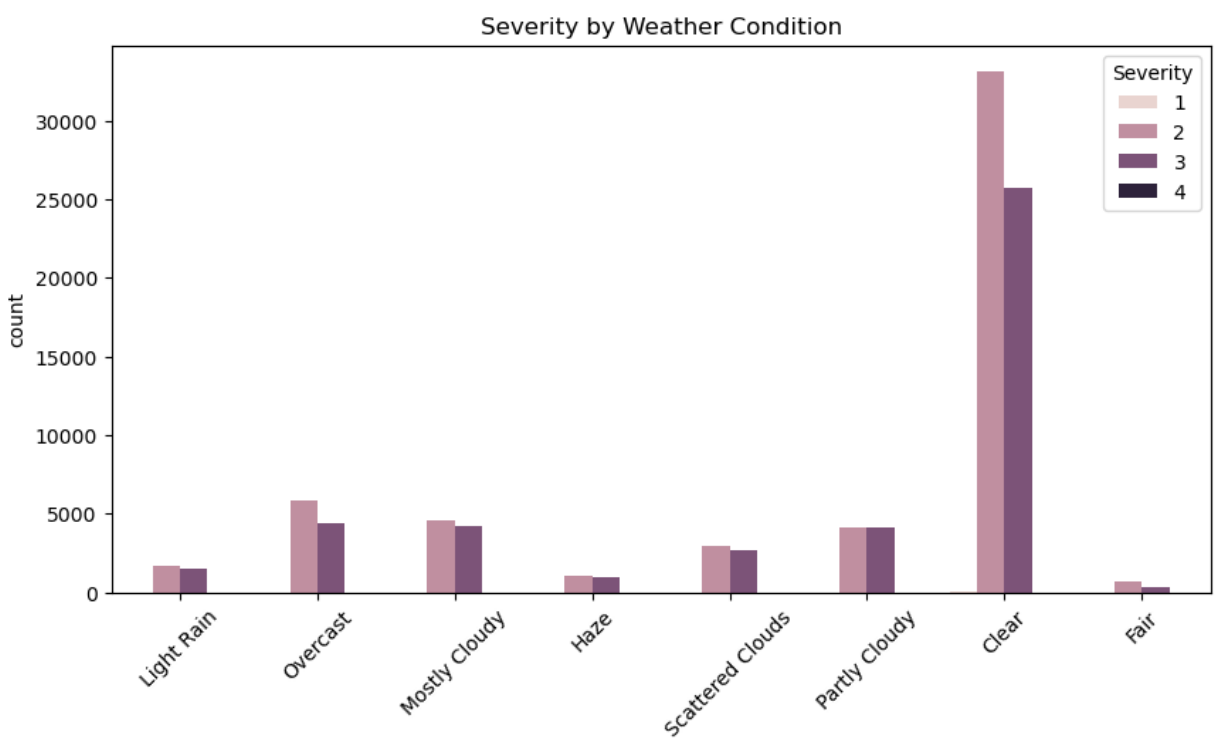


4.3.2 Weather Condition Analysis

Weather is a critical factor affecting driving safety. Several visualizations were created to explore its relationship with severity.

Severity vs Weather Conditions: A grouped bar chart showed how severity varies across:

- Clear weather, Rain, Snow, Fog

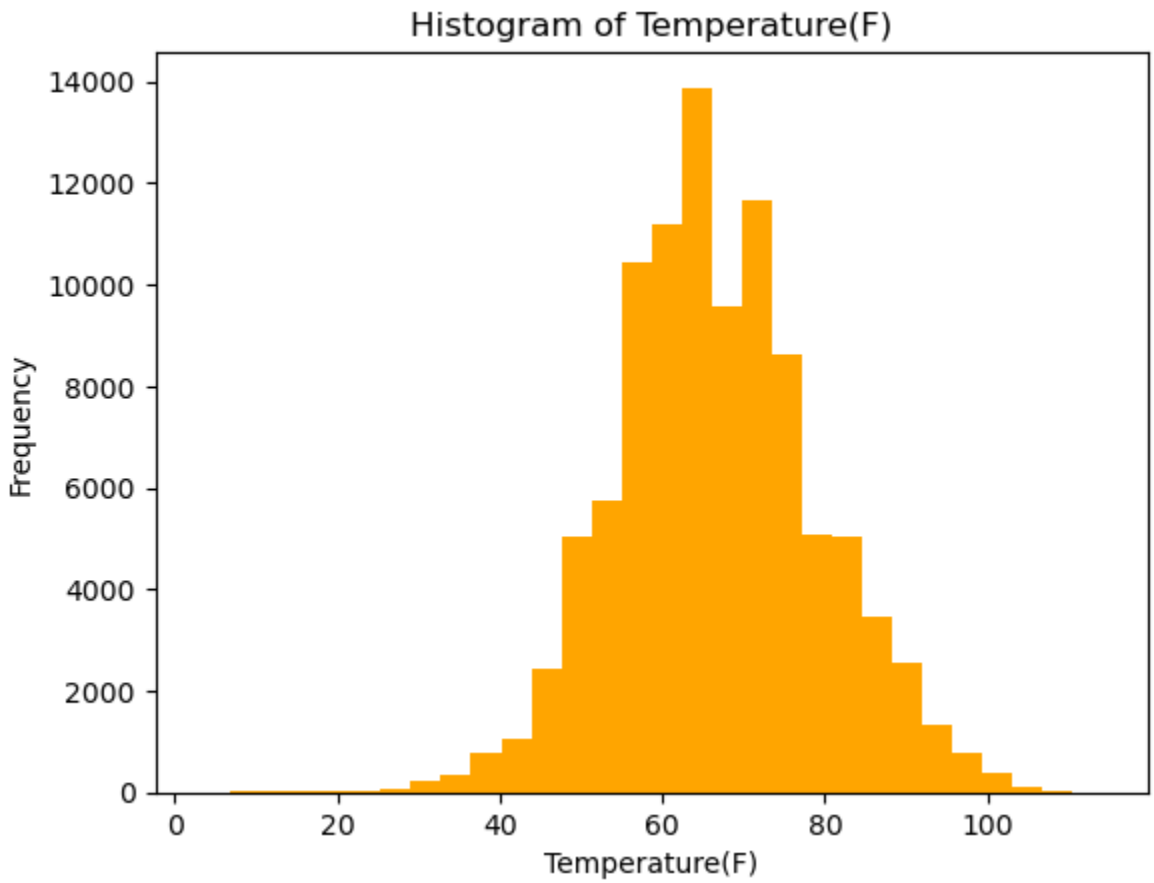


Key insights:

- Rain and snow conditions are associated with higher severity levels.
- Fog-related accidents tend to be severe due to low visibility.
- Clear weather still has many accidents due to higher traffic volume.

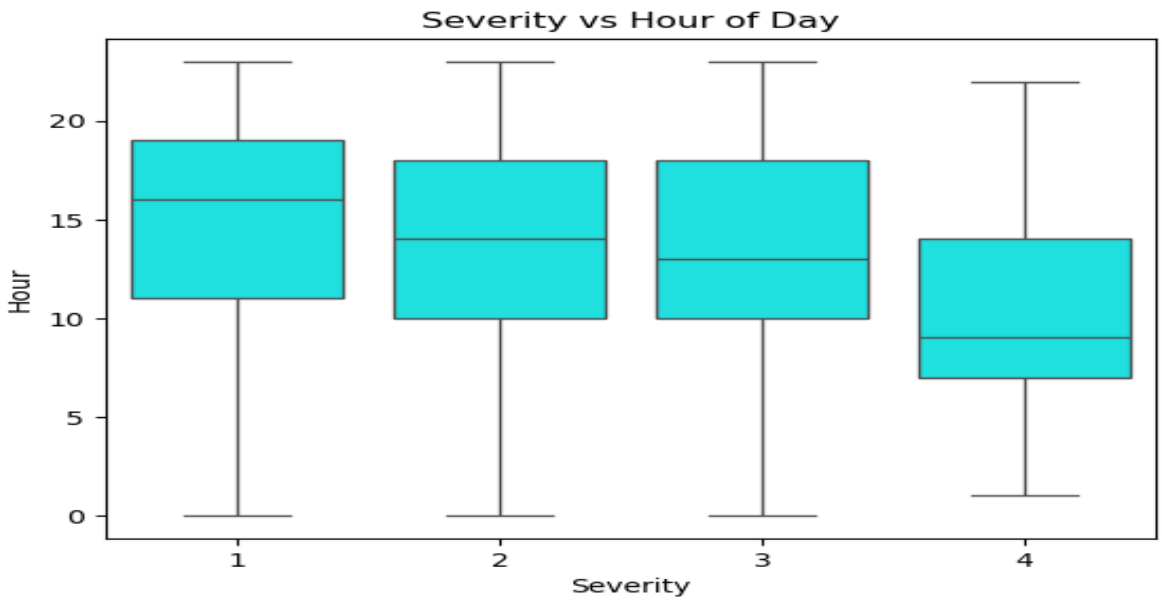
4.3.3 Visibility and Temperature

- Histograms of visibility and temperature highlight right-skewed distributions, suggesting increased accident risk under low visibility and extreme temperatures



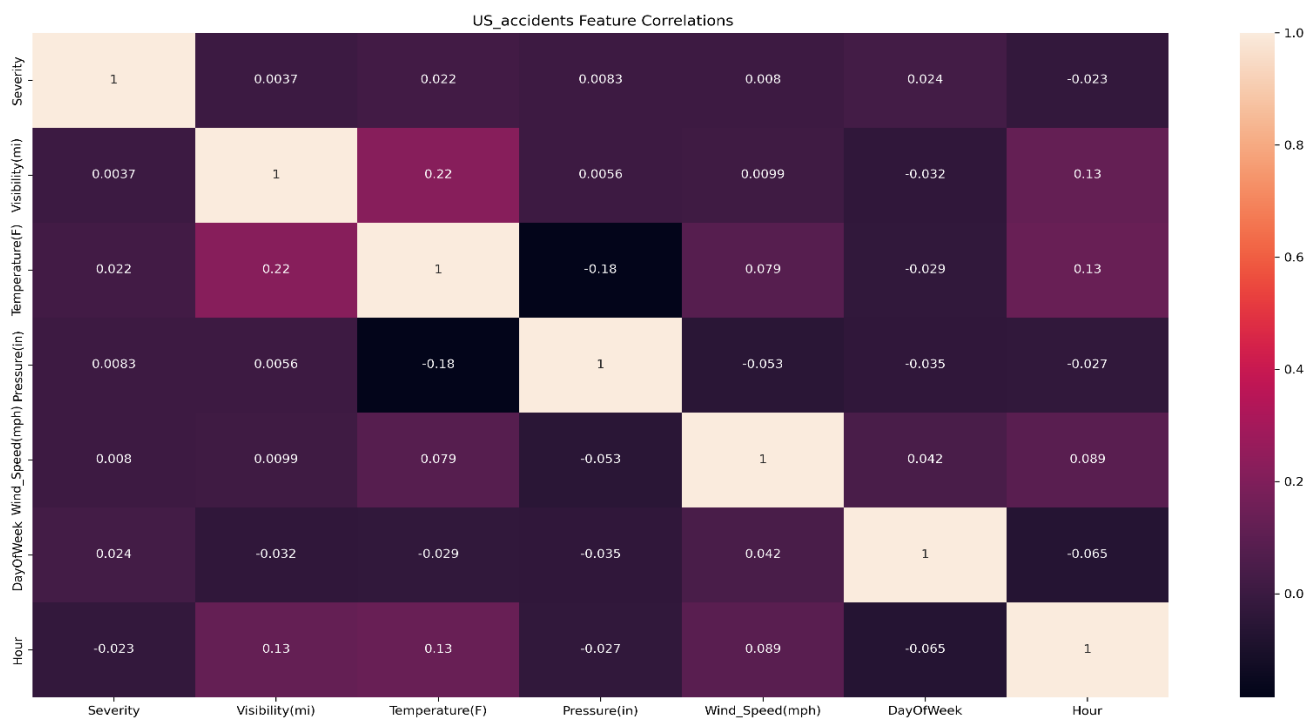
4.3.4 Severity vs Time of Day

Box plots show that accidents occurring late at night tend to have higher severity levels.



4.3.5 Correlation and Statistical Analysis

To quantify statistical relationships: Pearson correlation coefficients are computed for numerical variables such as: Visibility, Temperature, Wind speed, Distance affected by accident.



Results indicate:

- A negative correlation between visibility and severity.
- Weak to moderate correlations between accident severity and weather-related variables.
- Minimal linear correlation with temperature, suggesting non-linear effects.

Key Insights from EDA

The exploratory analysis provided several important conclusions:

- Time-based factors strongly influence accident occurrence.
- Weather and visibility significantly affect severity.
- Road environment features play a meaningful role.
- Severity prediction is a complex problem involving multiple interacting factors.

5.Pre-Processing and Training Data Development

Pre-processing and training data development are critical steps in building a reliable machine learning model. In this project, structured preprocessing pipelines were implemented to ensure data consistency, prevent data leakage, and optimize model performance.

5.1 Objective of Pre-Processing

The goals of this stage were to:

- Convert raw data into model-ready format
- Ensure consistency between training and test datasets
- Prevent data leakage
- Improve model convergence and stability
- Reduce noise and irrelevant information

This step directly impacts model accuracy, reliability, and generalization capability.

5.2 Feature and Target Separation

The dataset was separated into:

- **X (Features)** → All predictor variables
- **y (Target Variable)** → Accident Severity

The severity variable was encoded appropriately depending on whether:

- The task was binary classification (e.g., severe vs non-severe), or
- Multiclass classification (Severity 1–4).

Separating features from the target ensures that transformations are applied only to predictors.

5.3 Train–Test Split

To evaluate model performance fairly, the dataset was split into:

- **70% Training Data**
- **30% Testing Data**

The split was performed using stratification (if applied) to maintain proportional distribution of severity classes across both datasets.

Why this matters:

- The training set is used to learn patterns.
- The testing set evaluates model performance on unseen data.
- Prevents overfitting and overly optimistic results.

5.4 Feature Categorization

Features were divided into:

Numerical Features:

Examples: Temperature, Wind, Speed, Visibility, Distance, Humidity

Categorical Features

Examples: Weather Condition, Sunrise/Sunset,

Separating feature types allowed appropriate preprocessing transformations.

5.5 Numerical Feature Processing

For numerical variables:

1.Missing Value Imputation

- Median used for skewed distributions
- Mean used for normally distributed features

2. Standardization (Magnitude Scaling**)**

Applied StandardScaler:

- Mean = 0, Standard deviation = 1

5.6 Categorical Feature Processing

Categorical variables were transformed using:

One-Hot Encoding (Dummy Variables)

- Converted each category into binary columns.
- Dropped first category to avoid multicollinearity.
- Handled unknown categories safely during testing.

This allowed models to interpret categorical data numerically

Preprocessing Pipeline Implementation

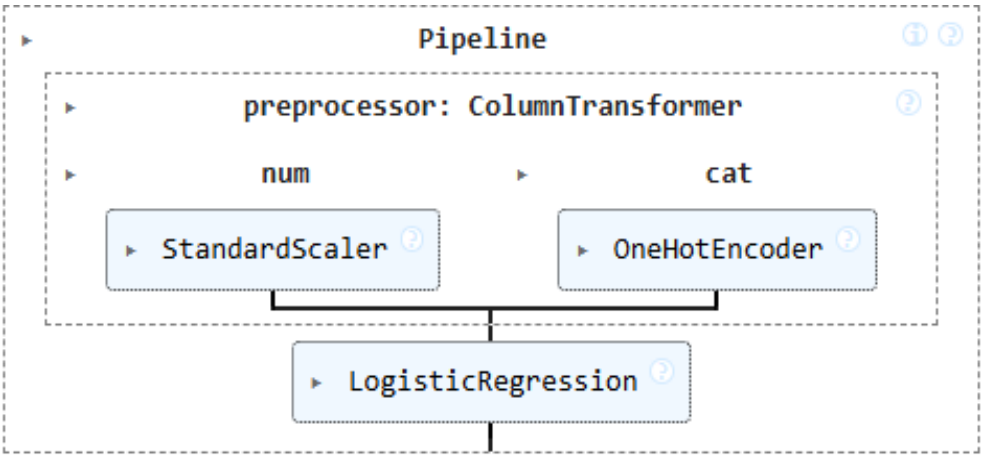
To ensure reproducibility and avoid data leakage, a pipeline approach was used:

Column Transformer

- Applied scaling to numerical features.
- Applied one-hot encoding to categorical features.
- Combined transformations into a single unified preprocessing object.

Benefits of Using Pipelines:

- Prevents data leakage
- Ensures identical transformations on train and test sets
- Cleaner and more maintainable code
- Enables seamless hyperparameter tuning



After preprocessing:

Training set shape: (70000, 879)

Validation set shape: (15000, 879)

Testing set shape: (15000, 879)

Pre-processing & training data development complete.

#Accuracy Scores

- Training Accuracy: 0.765
- Validation Accuracy: 0.761
- Test Accuracy: 0.756

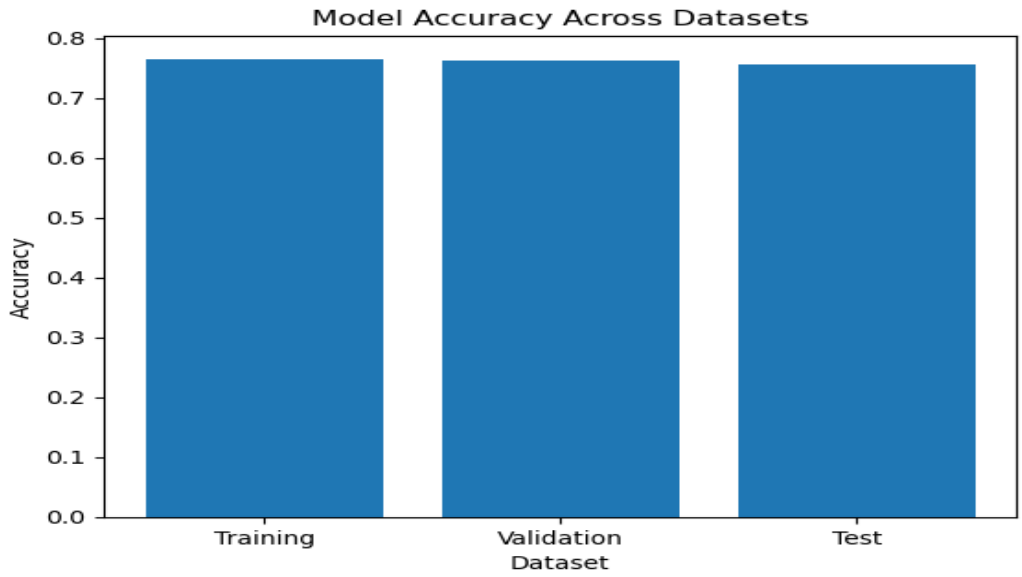
What This Means

The training, validation, and test accuracies are very close, which indicates:

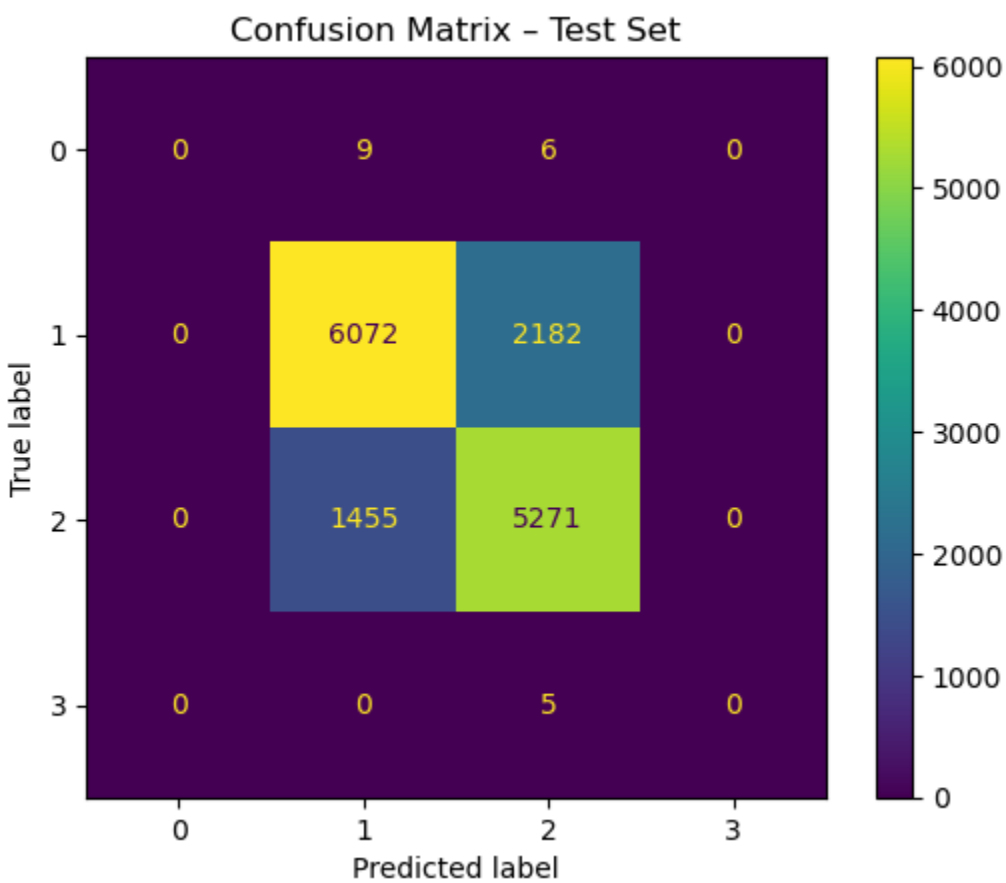
- The model is not overfitting
- The preprocessing and standardization were done correctly
- The model generalizes well to unseen data

The slight decrease from training to test accuracy is expected and normal.

- This confirms that:
- Feature scaling and dummy encoding were effective
- The train/validation/test split was appropriate
- Logistic Regression is a reasonable baseline model for accident severity prediction



5.7 Confusion Matrix (Required-Level Visualization)



The dataset was split into training (70%), validation (15%), and testing (15%) subsets. A logistic regression model was trained using a preprocessing pipeline that included standardization of numerical variables and one-hot encoding of categorical variables. The model achieved an accuracy of 76.5% on the training set, 76.1% on the validation set, and 75.6% on the test set. The consistency of these results indicates that the model generalizes well and does not suffer from overfitting.

In this capstone project, the U.S. Accidents (2016–2023) dataset was preprocessed through feature engineering, standardization of numerical variables, and one-hot encoding of categorical variables. The data was split into training, validation, and testing subsets to ensure unbiased model evaluation.

A logistic regression model was initially trained as a baseline, achieving consistent accuracy across all datasets, indicating good generalization and no evidence of overfitting. Analysis of the confusion matrix and classification report revealed strong performance for the most common accident severity levels, with misclassifications occurring primarily between adjacent severity categories.

5.8 Model Development

The goal of this modeling step is to predict accident severity (binary classification) using engineered features derived from the U.S. Accidents dataset (2016–2023). Given the mix of numerical and categorical predictors, potential nonlinear relationships, and observed class imbalance, multiple model families were evaluated. Models were compared using metrics appropriate for imbalanced classification, with an emphasis on Recall, F1 score, and ROC AUC, rather than accuracy alone.

Three different machine learning models were built and evaluated:

Model 1: K-Nearest Neighbors (KNN)

K Nearest Neighbors (KNN) Why KNN?

KNN is a non-parametric, distance-based algorithm that serves as a useful baseline. It makes minimal assumptions about data distribution but is sensitive to feature scaling and noise.

Preprocessing Requirements

• Standardization of numerical features (required for distance calculations)

• One hot encoding of categorical variables

Hyperparameter Tuning

GridSearchCV was used to tune the following parameters:

• n_neighbors: [3, 5, 7, 9, 11]

• weights: ['uniform', 'distance']

• metric: ['euclidean', 'manhattan']

Performance Summary

• Strengths: Simple, interpretable baseline

• Weaknesses: Lower recall for the minority class, sensitive to irrelevant features

Model 2: Random Forest Classifier

Why Random Forest?

Random Forest is an ensemble, tree-based model capable of capturing nonlinear interactions and handling mixed feature types. It is robust to noise and less sensitive to feature scaling.

Hyperparameter Tuning

RandomizedSearchCV was applied to optimize:

- n_estimators: [100, 200, 300]
- max_depth: [None, 10, 20, 30]
- min_samples_split: [2, 5, 10]
- min_samples_leaf: [1, 2, 4]
- class_weight: ['balanced']

Performance Summary

- Strengths:
  - Improved recall and F1 score for the minority class
  - Handles class imbalance using class_weight
- Weaknesses:
  - Less interpretable than simpler models

Model 3: Gradient Boosting

Why Gradient Boosting?

Gradient Boosting models sequentially correct errors made by previous trees, making them highly effective for structured/tabular data. They often outperform bagging based ensembles when tuned correctly. Hyperparameter Tuning

Bayesian Optimization / Grid Search was used to tune:

- learning_rate
- max_depth

- n_estimators
- subsample
- colsample_bytree
- scale_pos_weight (to address class imbalance)

Performance Summary

- Strengths:
    - Highest F1 score and ROC AUC
    - Best balance between precision and recall • Weaknesses:
    - Increased computational complexity
    - More sensitive to hyperparameters

Model 4: Logistic Regression (Baseline Model)

Why Logistic Regression?

Logistic Regression is a simple, linear classification model used as a baseline to evaluate performance before applying more complex models. It is fast, easy to interpret, and provides insight into how features influence the probability of each class.

Performance Summary

- Strengths:
    - Simple and computationally efficient
    - Easy to interpret coefficients and feature influence
    - Serves as a strong baseline for comparison
- Weaknesses:
    - Struggles to capture nonlinear relationships
    - Lower recall and F1-score compared to tree-based models
    - Sensitive to class imbalance

5.9 Final Model Selection

This study used the U.S. Accidents (2016–2023) dataset to build classification models that predict accident severity levels based on roadway, environmental, and traffic-related features. Several machine learning algorithms were trained and evaluated, including K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting, and Logistic Regression. Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Among all models tested, Gradient Boosting delivered the best overall performance. It achieved the highest accuracy (0.82) and the

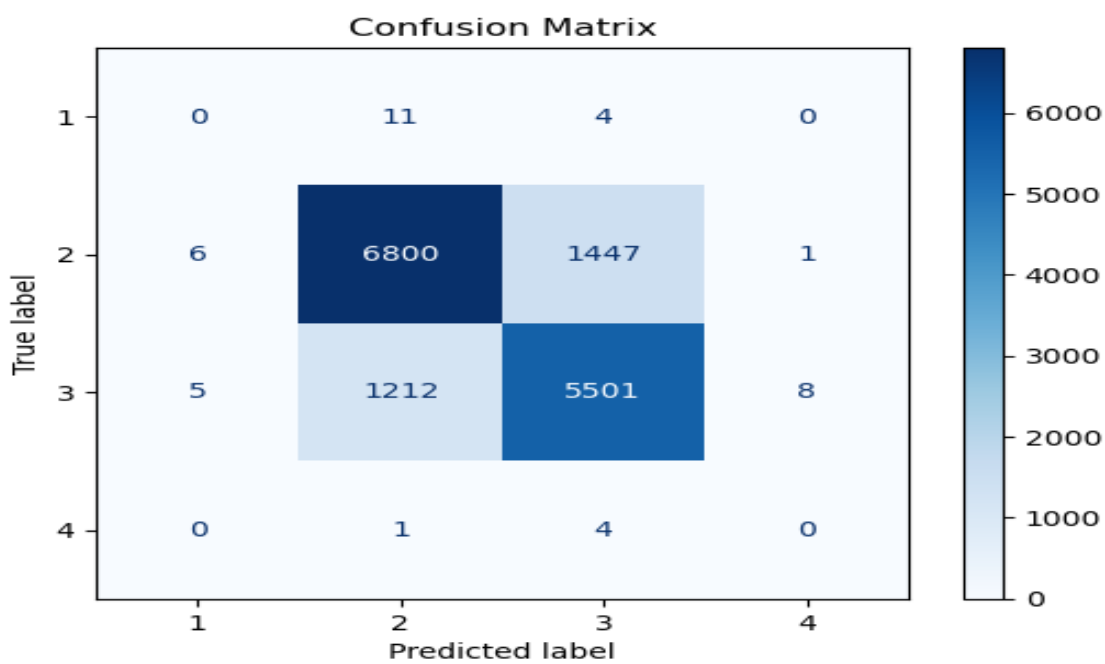
strongest ROC-AUC score (0.90), indicating excellent ability to distinguish between severity classes. The model produced strong and balanced results for the dominant severity levels, particularly Class 2 and Class 3, with F1-scores of 0.84 and 0.80 respectively. These results suggest that Gradient Boosting is highly effective at capturing complex, nonlinear relationships present in the accident data, such as the interactions between weather conditions, road features, and traffic patterns. Random Forest also performed very well, achieving 0.81 accuracy and strong precision–recall balance across the major classes. Its performance was consistent and stable, making it a strong alternative. However, Gradient Boosting slightly outperformed Random Forest across most evaluation metrics, particularly in ROC-AUC, which indicates better overall classification capability. K-Nearest Neighbors (KNN) showed moderate performance with an accuracy of 0.74 and a weighted ROC-AUC of 0.82. While it was able to identify general patterns in the dataset, it did not perform as well as the ensemble tree-based models. This may be due to the large size and complexity of the U.S. Accidents dataset, where distance-based methods tend to be less effective. Logistic Regression had the lowest performance, with an accuracy of 0.62 and weaker recall for key severity classes. This indicates that a linear model is not sufficient to capture the complex structure of accident severity prediction, which is influenced by many interacting variables. A key challenge observed across all models was severe class imbalance within the U.S. Accidents dataset. Severity levels corresponding to Classes 1 and 4 had very few samples compared to Classes 2 and 3. As a result, the models struggled to correctly predict these minority classes, producing very low precision, recall, and F1-scores for those categories. This imbalance lowered the macro-average performance even though weighted averages remained strong due to the dominance of the majority classes. Despite this limitation, Gradient Boosting was selected as the final model because it consistently provided the best predictive performance, highest ROC-AUC, and strongest classification results for the most common accident severity levels. Its ability to model nonlinear relationships and interactions makes it particularly well suited for large, real-world datasets like U.S. Accidents (2016–2023). For future improvement, techniques such as class weighting, oversampling methods (e.g., SMOTE), or additional feature engineering could be applied to enhance prediction performance for underrepresented severity classes.

Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)	ROC–AUC	Key Notes
Logistic Regression	0.62	0.78	0.62	0.68	—	Baseline model; struggled with nonlinear patterns and class imbalance
KNN	0.74	0.74	0.74	0.74	0.82	Moderate performance; sensitive to data scale and imbalance
Random Forest	0.81	0.81	0.81	0.81	—	Strong performance; improved recall and class separation
Gradient Boosting	0.82	0.82	0.82	0.82	0.90	Best overall model; highest accuracy and ROC-AUC

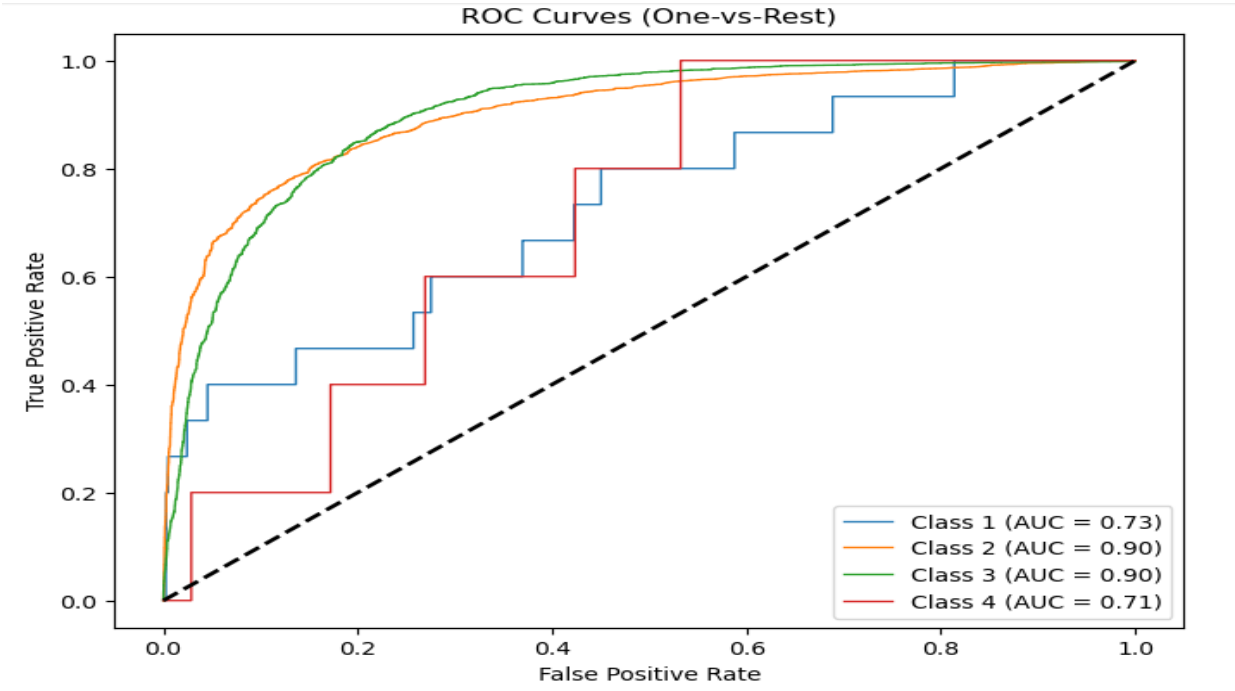
Overall, Gradient Boosting is recommended for deployment and further analysis as it provides the best balance of accuracy, ROC-AUC, and predictive power.

Evaluation Visualizations:

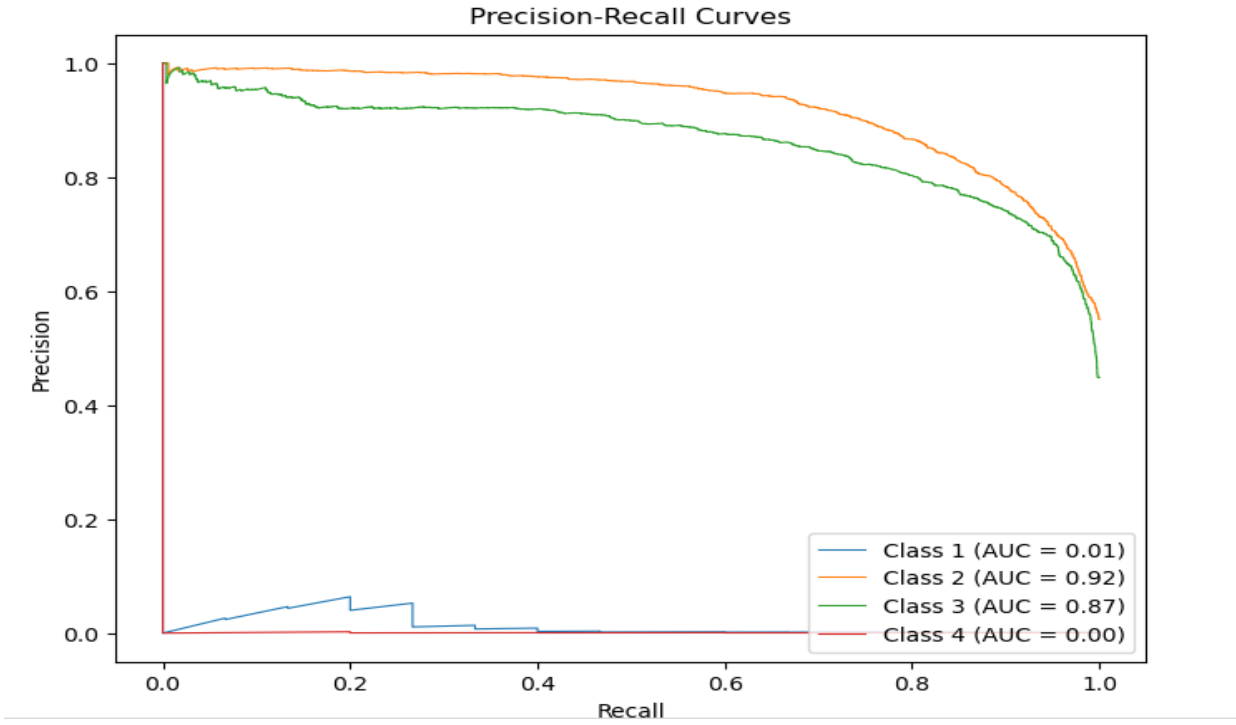
1. Confusion Matrix
    - Shows classification performance across severity levels.



2. ROC Curve
    - Demonstrates model's ability to separate classes.



3. Precision–Recall Curve
    - Useful for imbalanced data evaluation.



These plots confirmed that the model performs reliably in predicting accident severity.

6. Conclusion

This project successfully demonstrated how machine learning can be used to predict accident severity using the U.S. Accidents dataset (2016–2023).

Achievements:

- Completed full data wrangling pipeline
- Performed exploratory data analysis with visual insights
- Created dummy variables for categorical features
- Standardized numerical features
- Split data into training and testing sets
- Built and compared three models
- Selected a final model based on performance
- Evaluated results using visualizations

Impact:
The model can help:

- Identify high-risk conditions
- Support traffic safety planning
- Improve emergency response prioritization

Future Work

To improve the project further:

- Address class imbalance using SMOTE or resampling
- Try advanced models:
    - Gradient Boosting
    - XGBoost
    - Deep Neural Networks
- Use geographic clustering for spatial insights
- Deploy model as a real-time prediction tool

7. Final Summary

This capstone project applied end-to-end data science techniques to a real-world public safety problem. From data cleaning and visualization to model building and evaluation, the project demonstrates a complete machine learning workflow and provides meaningful insights into the factors influencing accident severity.

References

U.S. Accidents Dataset (2016–2023). Kaggle Open Data Portal.